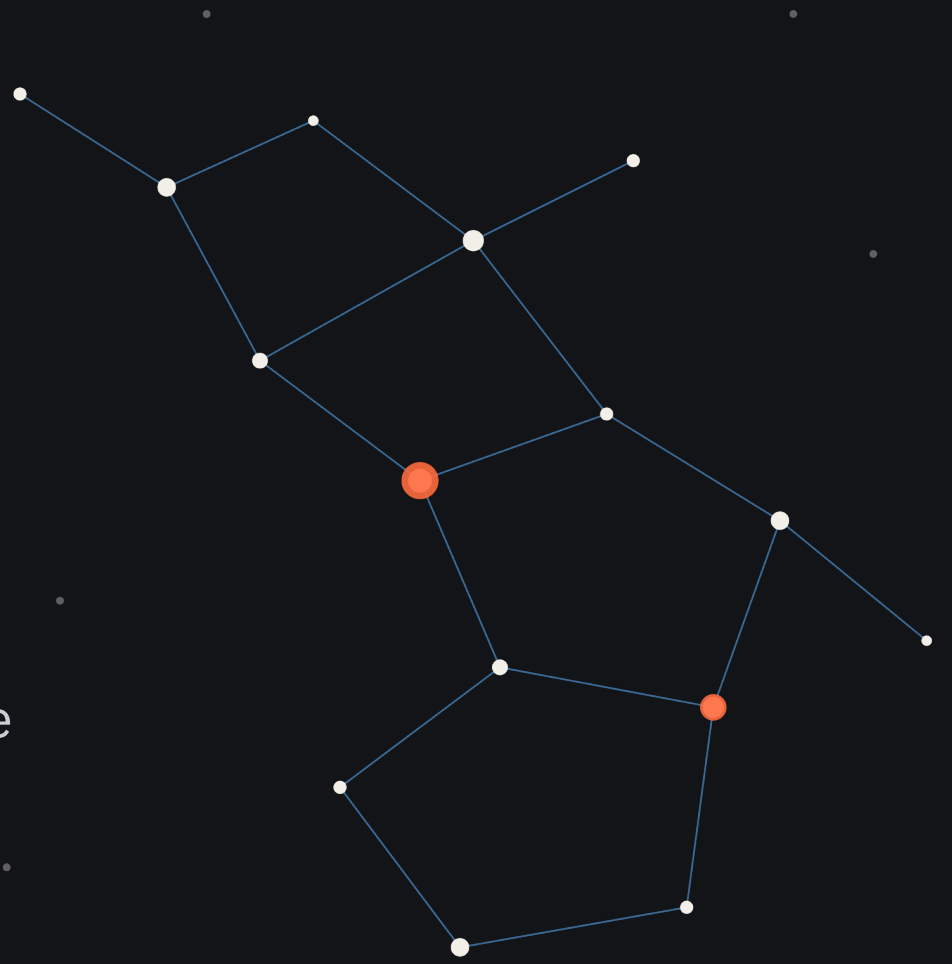
TOUCHDESIGNER MASTER CLASS 2

Points, Bodies, Systems

Point Operators (POPs) and how to design an interactive system around them



Jojo Zhou · Norwich University of the Arts · Creative Computing

THE DAY · 6 HOURS

Three short labs, then your own prototype

Time	Block	Mode
9:30	Recap and bridge from Master Class 1	Talk
10:00	1 · POP foundations	Talk + demo
10:30	Lab 1 · Image to points, rebuilt with POPs	Lab
11:30	2 · Making points move	Demo
12:00	Lab 2 · Living points	Lab
12:30	Break	—
13:30	3 · Designing interaction	Talk
14:00	Lab 3 · Body to points	Lab
16:00	Show and tell	Crit

FROM MASTER CLASS 1

Last time we moved data around the points. Today we work on the points.

MASTER CLASS 1

Instancing

Image → TOP to CHOP → channels → Geometry COMP instance page. Each step converts the data into a new format.



MASTER CLASS 2

POPs

Image → TOP to POP → points that carry position and colour, on the GPU. They stay points the whole way through.

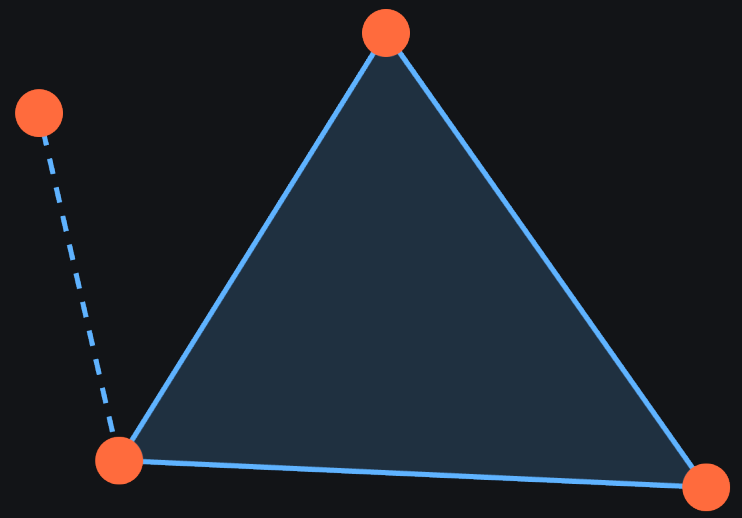
What is a POP?

A **Point Operator** creates or changes 3D data on the GPU. Every POP holds three lists:

Points carry attributes: position, colour, anything you name.

Primitives say how points join: point, line, line strip, triangle, quad.

Vertices are the links from a primitive back to its points.



orange = points · blue = primitives

Attributes are the data. Name one and it exists.

Attribute	Holds
P	Position (x, y, z)
Color	RGBA colour
N	Normal direction
PartVel	Particle velocity
_PointI	Point index (read-only)
Excite	Any name you invent

Three habits

Look: turn on attribute display in the POP viewer, or wire a POP to DAT to read the numbers.

Swizzle: P.y or P.yxz picks and reorders components.

Trust the pass-through: a POP only stores what it changes, so long chains stay cheap.

SOPs and POPs

	SOPs	POPs
Runs on	CPU	GPU compute shaders
Scale	Thousands of points	Millions of points
Data	Fixed attribute set	Any attribute you name
Instancing	Often via CHOPs or DATs	Read attributes directly
Still better for	Booleans, NURBS, meshes	—

Sources: Derivative, Learning About POPs; Interactive & Immersive HQ, POPs in TouchDesigner FAQ. Derivative asks for 4 GB graphics memory minimum, 8 GB recommended.

Over 100 POPs, four verbs

Make

Create or import points

Grid
Sphere
Pattern
TOP to POP
CHOP to POP
Point File In

Shape

Change attributes

Transform
Math
Noise
Field
Random
Lookup Texture

Move

Carry state frame to frame

Feedback
Particle
Force Radial
Trail
Accumulate

Connect

Build lines and surfaces

Proximity
Line
Line Thick
Copy
Polygonize

DEMO · 10 MIN

Last week’s patch, in four operators



TOP to POP turns pixels into points. Its *Height (R)* mode pushes points by brightness; *Color (RGBA)* carries the image’s colour.

No CHOPs, no instance page. Point size comes from the PointScale attribute.

LAB 1 · 40 MIN

Image to points, rebuilt

1. Camera or movie in, reduced to about 256 × 256
2. TOP to POP: depth from brightness
3. Carry the image’s colour onto the points
4. Add a Noise POP and animate it over time
5. Render: Geometry COMP, Camera, Render TOP

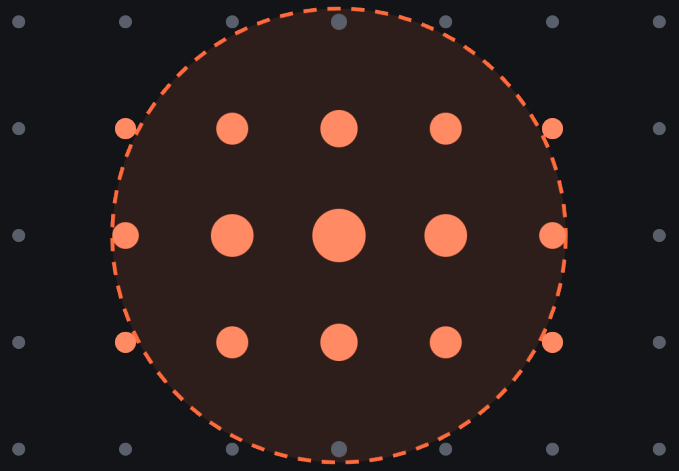
Check and stretch

Compare cook time with your Master Class 1 patch.

Switch Connectivity from points to triangles. What happens?

Save as lab1.toe. You will reuse it.

A field decides *where* something happens

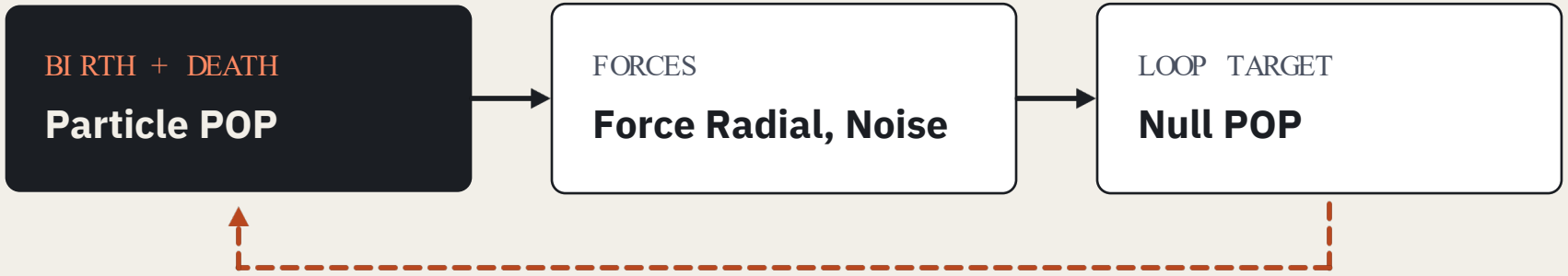


Field POP writes a Weight from 1 inside a shape to 0 outside, with a soft edge.

Then let Weight scale an effect: on the Noise POP's **Map** page, drive Amplitude from Weight.

Move the field's centre with a hand, and the hand now has a zone of influence.

Memory: the particle loop



The Particle POP is built on the Feedback POP: its **Target** is the Null at the end, read back one frame later.

Only the Particle POP adds or removes points. Forces add to PartForce; velocity and position follow.

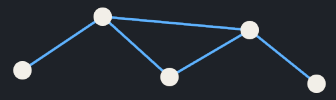
2 · MAKING POINTS MOVE

Lines turn movement into drawing



Trail POP

Keeps the last N seconds of points and joins them into rows: a gesture leaves a mark.



Proximity POP

Joins points closer than a distance: a network that forms and breaks as things move.



Line Thick POP

Gives lines a real width, so a trail renders as a ribbon rather than a hairline.

LAB 2 · 40 MIN

Living points

1. Emit particles from your Lab 1 points
2. Give them a life span and a little drag
3. Add one force: Force Radial or Noise
4. Add a Field POP and limit an effect to it
5. Draw with them: Trail or Proximity

Check and stretch

Move the field with an LFO CHOP. It is a stand-in for a hand.

Colour by age: map PartAge to Color.

Save as lab2.toe.

3 · DESIGNING INTERACTION

What does the body do, and what does the system say back?

You can now make points do almost anything. The hard part is deciding which movement should cause which change, and whether a person can feel that it did.

Three questions for every interaction

Do

How does a person act on it? A button is discrete and precise. A gesture is continuous and expressive.

Feel

What comes back to the senses? Is the feedback immediate, and does it leave room for interpretation?

Know

What does the person learn? Step by step (a path), or a picture of how the whole system behaves (a map)?

After Bill Verplank, Interaction Design Sketchbook (Stanford CCRMA course notes)

Mapping is the design

Shapes of a mapping

Mapping	Example
One to one	Hand height → field size
One to many	Hand speed → noise, colour, trail length
Many to one	Two hands' distance → force strength

One to one is easy to read. Many to one tends to feel more like an instrument.

Anatomy of one interaction

Trigger: what starts it
Rules: what the system does
Feedback: how the person sees it
Loops: what changes over time

Hunt & Wanderley (2002), Mapping performer parameters to synthesis engines, Organised Sound 7(2); Saffer (2013), Microinteractions

Four rules from my own practice

Legible

A few clear mappings a person can learn

Fast

Responsiveness before more features

Felt

Output the body can sense and answer

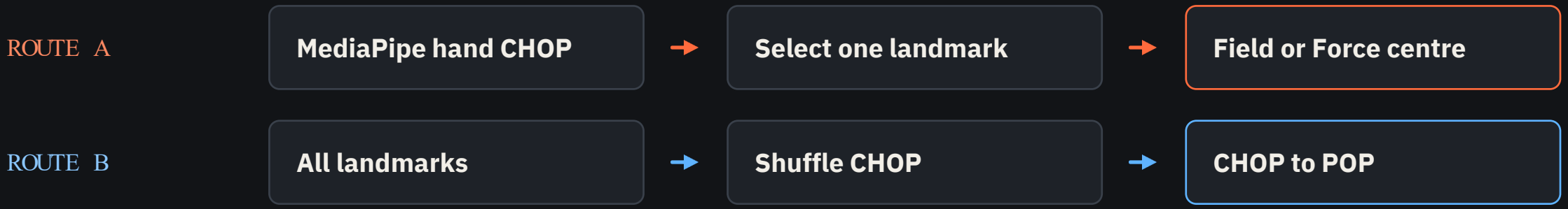
Bounded

Some surprise, without losing control

“Set the stage, then shut up and listen.” — Tom Igoe

DEMO · 10 MIN

From MediaPipe to points: two routes



A: the hand as a control. Export x and y to a parameter. Simple, and works for one point of contact.

B: the hand as geometry. CHOP to POP makes one point per sample, so landmarks become points you can join, trail or emit from.

LAB 3 · 40 MIN

Body to points

1. Open lab2.toe and add the MediaPipe component
2. Route A: replace the LFO with your index fingertip
3. Route B: turn all 21 hand landmarks into points
4. Join them with Proximity, or trail them
5. Write your Do / Feel / Know in three sentences

Test it on someone

Swap seats. Say nothing. Can your partner find the rule in 30 seconds?

Write down what they tried first.

STUDIO SPRINT · CHOOSE ONE BRIEF

Four briefs

A · The mirror that remembers

A body leaves traces that fade slowly. Camera or MediaPipe, Trail, Feedback.

B · Sound you can see

Audio from Ableton or Max over OSC shapes a particle field. Fields, forces.

C · The hidden portrait

An image as points, revealed only where a hand passes. TOP to POP, Field.

D · Out of the screen

Points drive something physical or another app: DMX, LEDs, or OSC out.

STUDIO SPRINT · 50 MIN

What to show at 16:00

The rule

At least **one** technique from Master Class 1, and at least **two** POP verbs from today (make, shape, move, connect).

Start from your Lab 3 file. Build the input with a placeholder first, then plug in the real one.

Show and tell: 2 minutes each

- A live demo, or a 30-second recording
- Your mapping, as one sketch
- Do / Feel / Know, three sentences
- One thing you would change

WHEN IT GETS SLOW

Keep it real -time

Measure first. Performance Monitor or the Probe palette shows which operator costs the most.

Stay on the GPU. Reading POP data back to the CPU (DATs, Python) can stall the frame.

Cap the count. Reduce input resolution and set a maximum number of points.

Watch the frame rate with a Perform CHOP on screen while you test with a person.

READING AND RESOURCES

Where to go next

POPs

Derivative · Learning About POPs

Derivative Curriculum · Lesson 109, POPs: Working with Points

POPs Examples package · start with Overview.toe

OP Snippets · Help menu in TouchDesigner

Interactive & Immersive HQ · SOPs to POPs

Designing interaction

Igoe · Making Interactive Art: Set the Stage, Then Shut Up and Listen

Verplank · Interaction Design Sketchbook

Levin · Computer Vision for Artists and Designers

Hunt & Wanderley · Mapping performer parameters to synthesis engines

Höök · Designing with the Body

TO TAKE AWAY

1 Points carry their own data.

2 A field says where; an effect says what.

3 Build against a placeholder, then plug in the body.

4 Test with someone who has never seen it.